YUANKAI TAO

Email: yuankaitao0909@gmail.com | Tel: +1 470 658 3699 | GitHub | Homepage

EDUCATION

Georgia Institute of Technology *Aug 2026 — Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 — Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Advanced Risk Management, Computational Finance Methods, Data Science for Finance

Zhongnan University of Economics and Law *Sep 2020 — Jun 2024*
B.S. in Management, School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

TECHNICAL SKILLS

Programming & Tools Python, SQL, MATLAB, MS Office, LaTeX, Git, Linux, GCP
Quantitative Methods Advanced Risk Management (VAR, GARCH, Greeks, Delta Hedging), Regression Analysis, Probability Theory, Time Series Analysis, Simulation
Libraries & Frameworks Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPy, Backtrader

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 — Jul 2026*
Quantitative Research, Intern *Quantitative Strategy Department*

• Built a dynamic **VaR/CVaR risk measurement and model evaluation framework** for copper futures, comparing 17 models across five volatility specifications (including GARCH and EWMA) and Normal, Student-t, Skewed-t, and Filtered Historical Simulation assumptions; used semi-variance slicing to identify long/short momentum risk exposures.

• Engineered an end-to-end commodity futures **statistical and machine learning modeling** pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained and ensembled LightGBM, XGBoost, CatBoost, and neural networks across 1- to 20-day horizons.

• Applied **rolling out-of-sample model validation** using statistical and trading metrics; implemented a monthly rebalanced strategy with 120-day rolling retraining that achieved a **1.89 Sharpe ratio**, **16.79% annualized return**, and **25.51% excess return** after transaction costs.

VariFlight Technology *Jul 2025 — Sep 2025*
Data Analyst, Intern *Data Development Department*

• Developed Python web-scraping workflows to collect public reports, webpages, and images from the Civil Aviation Administration of China, supporting structured **data management** and downstream analytical use.

• Applied OCR and table-extraction tools, including pytesseract and Camelot, to convert **600+ unstructured PDF reports and 500+ webpages** into standardized CSV datasets for local LLM training, improving data quality and repeatability.

Ernst & Young *Dec 2022 — Feb 2023*
Assurance, Intern *Financial Service Office (FSO)*

• Supported the audit of intercompany and investment transactions for Hyundai Motor Finance Company; performed **variance analysis, risk assessment, and exception investigation** to identify and document significant deviations.

SELECTED PROJECTS

UCSD / FMSbonds, Inc. 2025 Capstone Project *Aug 2025 — Dec 2025*

Team Lead | High-Performance Bond Pricing Engine Optimization

• Developed a municipal bond **risk-screening and control framework** covering credit, structural, sector, tax, interest-rate, and data-integrity risks to support disciplined, data-driven risk identification.

• Designed caching for high-frequency cash-flow queries, eliminating redundant object generation and reducing per-bond solving time from 1.17 ms to 0.13 ms (**9× speedup**).

• Reimplemented the yield-to-worst numerical solver with Numba JIT, reducing runtime from 1.57 ms to 7.5 μs per bond (**210× speedup**) and enabling scalable portfolio analytics.

Kaggle Jane Street Market Prediction Competition *Silver Medal*

• Built a memory-efficient Polars/PyTorch pipeline for **47M intraday observations** across 1,669 trading days; engineered lagged, rolling, and temporal features and deployed online inference through Kaggle's streaming evaluation API.

• Implemented a parameter-efficient TabM regressor with a **32-member shared-parameter ensemble**; one-hot encoded three categorical features and improved cross-validation performance by **0.004**.

• Contributed to the baseline training and final ensemble of XGBoost, MLP, Ridge, and TabM, helping the team place **94th of 3,757 teams**.

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 — Jan 2026*

The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Engineered a **large-scale data pipeline** (Python, PostgreSQL, PostGIS) for 76 million spatial observations; managed a **964.5M-record, 100GB+ SQL database** on Google Cloud Platform and UNC Longleaf HPC using Python and DuckDB.

• Designed parallelized spatial computation and property-specific data controls across multiple distance ranges, delivering approximately **50× computational speedup** for scalable analytical workflows.

• Executed complex geospatial queries with time-window and property-level constraints to produce reproducible population statistics for quantitative research.

CERTIFICATIONS & HONORS

Deep Learning Specialization (5-Course Certificate) — Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate) — Coursera, UC San Diego

Beta Gamma Sigma (Top 20% of graduating class) — UCSD, *2024 — 2025*

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%) — *2024*

Special Scholarship in Innovation and Entrepreneurship — ZUEL, *2022 — 2023*